

Annealed Low Carbon Co-28Cr-6Mo Alloy

Annealed R31537 cobalt is R31537 cobalt in the annealed condition. It has the lowest strength and highest ductility compared to the other variants of R31537 cobalt. The graph bars on the material properties cards below compare annealed R31537 cobalt to other cobalt alloys (top) and the entire database (bottom). A full bar means this is the highest value in the relevant set. A half-full bar means it's 50% of the highest, and so on.

Mechanical Properties

Elastic (Young's, Tensile) Modulus

220 GPa32 x 10⁶ psi

Elongation at Break

23 %

Fatigue Strength

310 MPa45 x 10³ psi

Poisson's Ratio

0.29

Reduction in Area

23 %

Rockwell C Hardness

25

Shear Modulus

87 GPa13 x 10⁶ psi

Tensile Strength: Ultimate (UTS)

1000 MPa150 x 10³ psi

Tensile Strength: Yield (Proof)

590 MPa85 x 10³ psi

Thermal Properties

Latent Heat of Fusion

320 J/g

Melting Completion (Liquidus)

1360 °C 2480 °F

Melting Onset (Solidus)

1290 °C 2360 °F

Specific Heat Capacity

450 J/kg-K 0.11 BTU/lb-°F

Thermal Conductivity

13 W/m-K 7.3 BTU/h-ft-°F

Thermal Expansion

13 µm/m-K

Electrical Properties

Electrical Conductivity: Equal Volume

2.0 % IACS

Electrical Conductivity: Equal Weight (Specific)

2.1 % IACS

Otherwise Unclassified Properties

Density

8.4 g/cm³ 530 lb/ft³

Embodied Carbon

8.1 kg CO₂/kg material

Embodied Energy

110 MJ/kg 46 x 10³ BTU/lb

Embodied Water

530 L/kg 63 gal/lb

Common Calculations

Resilience: Ultimate (Unit Rupture Work)

200 MJ/m³

Resilience: Unit (Modulus of Resilience)

780 kJ/m³

Stiffness to Weight: Axial

15 points

Stiffness to Weight: Bending

24 points

Strength to Weight: Axial

33 points

Strength to Weight: Bending

26 points

Thermal Diffusivity

3.4 mm²/s

Thermal Shock Resistance

24 points